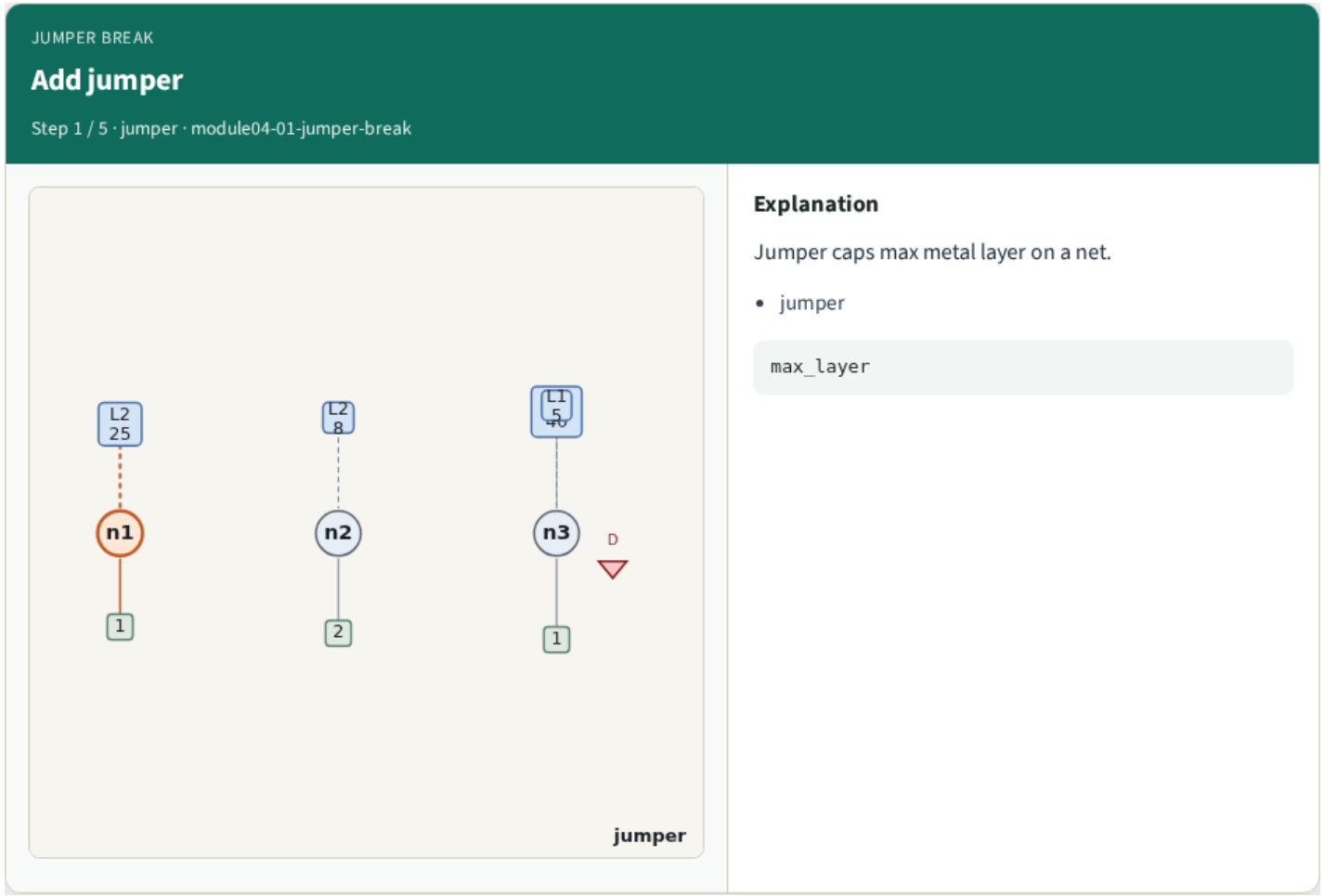
Why jumper

A jumper on lower metal breaks antenna path , drops high-layer metal

The idea

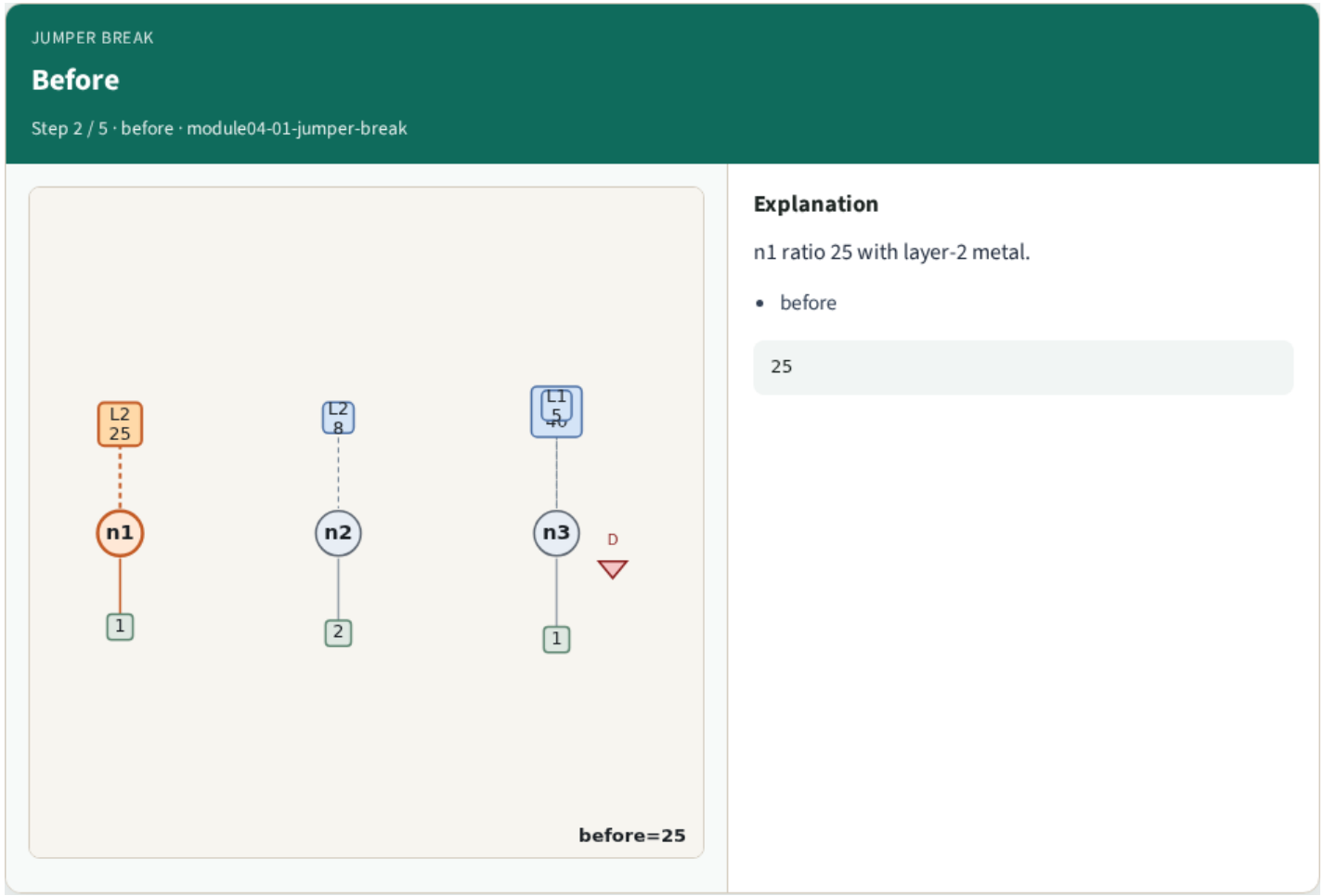
- GOLDEN: jumper on n1 max_layer=1 → ratio 0

Add jumper



Add jumper

Before



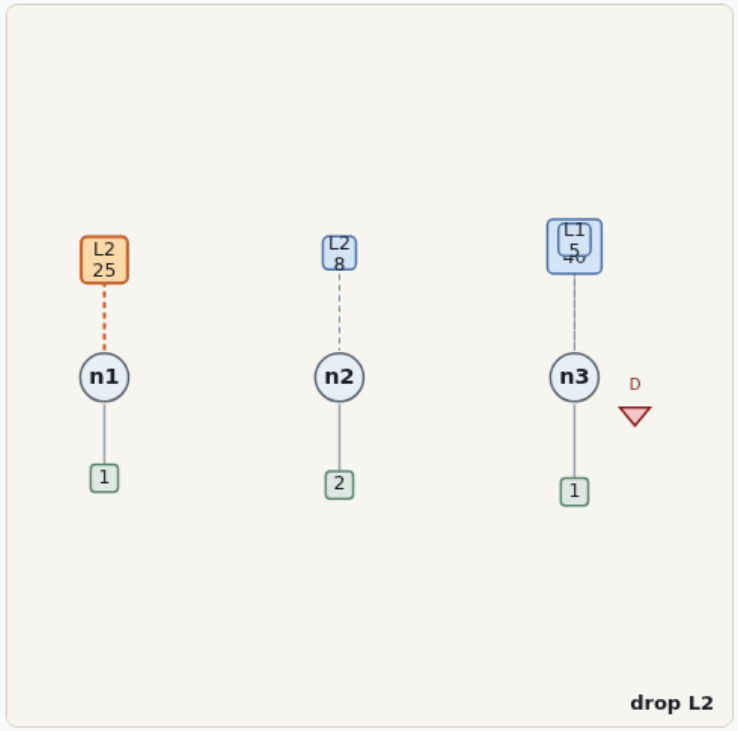
Before

Cap L1

JUMPER BREAK

Cap L1

Step 3 / 5 · cap · module04-01-jumper-break



Explanation

max_layer=1 drops m1 (layer 2).

- cap

```
L1
```

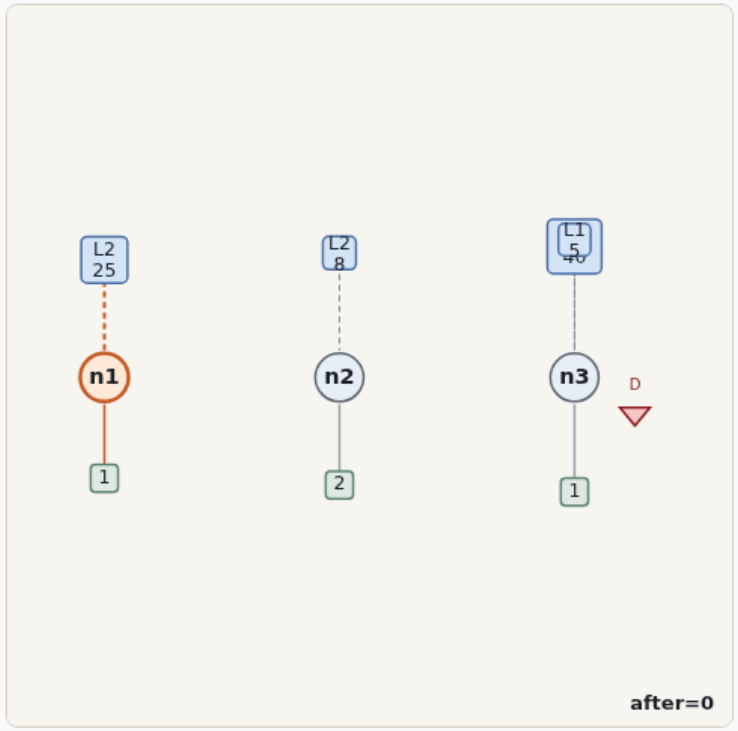
Cap L1

After

JUMPER BREAK

After

Step 4 / 5 · after · module04-01-jumper-break



Explanation

n1 metal area 0 → ratio 0.

- after

```
0
```

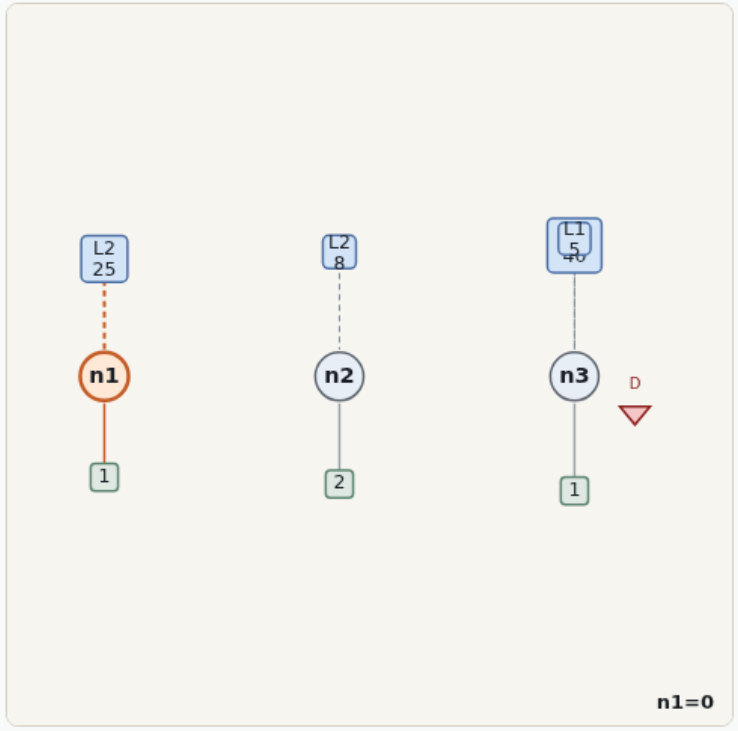
After

Golden

JUMPER BREAK

Golden

Step 5 / 5 · golden · module04-01-jumper-break



Explanation

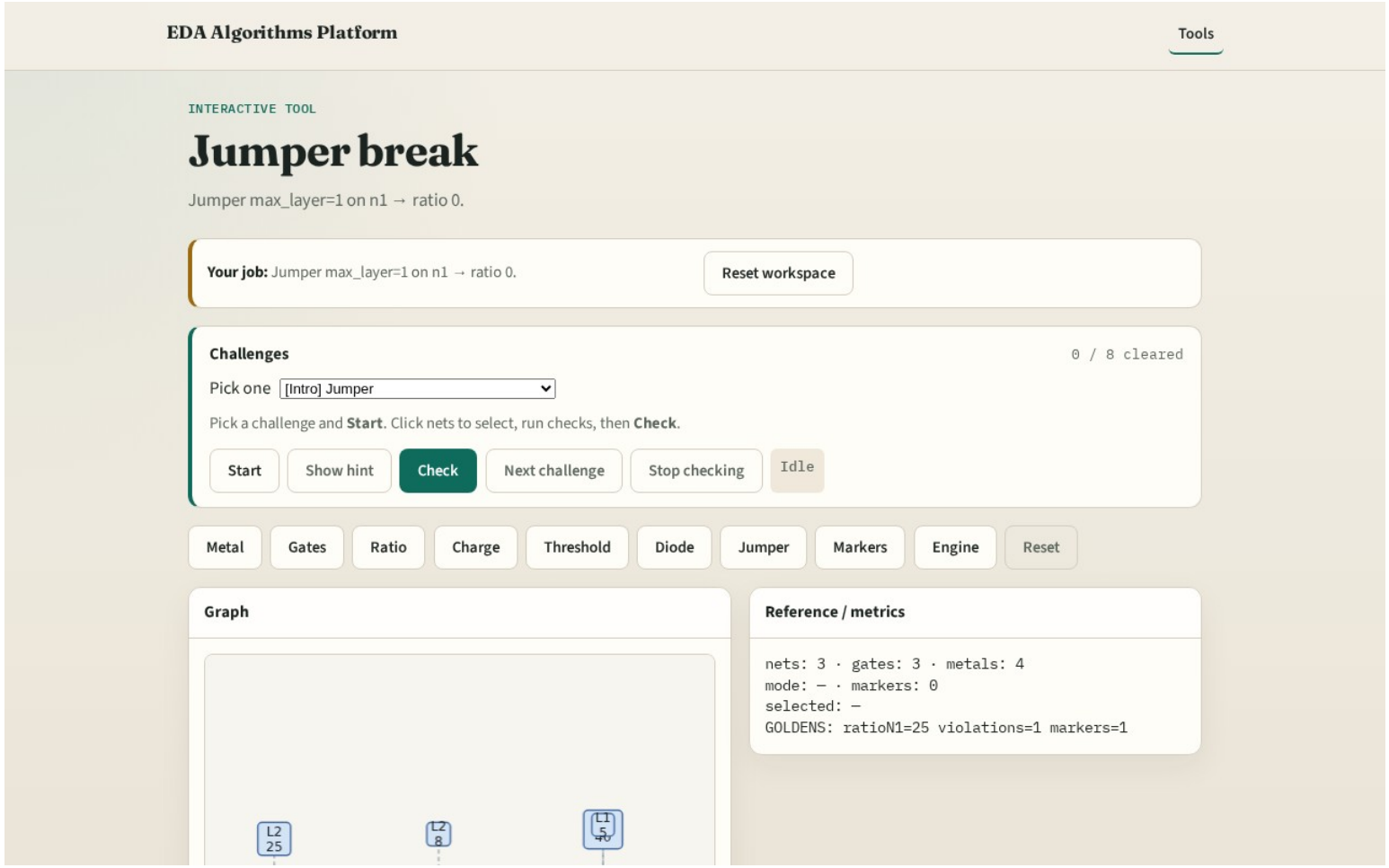
GOLDEN: jumper_n1_ratio=0.

- GOLDEN

```
0
```

Golden

Browser lab track



Browser lab starter

Implement track

- Call jumper_effect(); assert after_ratio==0

Pitfalls

- Not filtering by layer
- Applying jumper to wrong net

Your turn

- Next: antenna markers